

Learner-State-Aware Generative User Interfaces: A Closed-Loop Control Architecture for Adaptive Learning Environments

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Abstract—Intelligent tutoring systems estimate what a learner knows but present a fixed interface, while generative user interfaces (GenUI) synthesize interfaces at runtime without any model of the learner. We propose a closed-loop architecture that couples the two. Bayesian Knowledge Tracing estimates mastery; a PID governor with anti-windup and a filtered derivative converts the mastery deficit into a structural complexity budget; a fast policy selects scaffolding affordances; a deterministic planner fixes the layout whose complexity is closest to the budget; and a foundation model writes the content of that layout as a JSON component tree, with a fallback when schema or budget checks fail. We evaluate the architecture in silico on 1,000 synthetic learners from three archetypes, replayed through four arms. The closed loop keeps the rendered interface within 0.012 of the governor's budget on average, against 0.131 for unconstrained GenUI and 0.123 for a rule-based baseline, with 3.5 times less jitter than unconstrained GenUI, which matches latent mastery slightly better but skips levels three times as often. Standard BKT saturates after long success runs, so a later streak of errors leaves the interface unchanged; a CUSUM lapse detector cuts overload in a forgetting cohort by 25%. In 300 single calls to the real model with the planner (not full learner sessions), 98.7% of screens were within budget, against 92% when the model also chose the layout. All claims concern algorithmic behavior; no learning gains are claimed.

Index Terms—generative user interfaces, Bayesian knowledge tracing, PID control, cognitive load, adaptive learning, simulation

I. INTRODUCTION

Intelligent tutoring systems (ITS) have long modeled learners precisely. Knowledge tracing maintains a probability that a learner has mastered each knowledge component [1], and tutors built on it have been deployed at scale in mathematics [2], [3]. The interface of such systems, however, is authored in advance: the learner model selects the next problem, not the form in which the problem is presented.

Generative user interfaces take the opposite position. A foundation model produces a task-specific interface for each request [4], including interactive courseware [5]. These systems are open loop: the model does not observe the learner's state, and nothing bounds how complex the generated interface becomes.

Cognitive load theory explains why this matters. The same material can help a novice and hinder an expert (the expertise reversal effect) [6], and guidance should fade as competence grows [7], [8]. An interface that adapts to the learner therefore needs both a learner model and a mechanism that holds the interface to a complexity appropriate to that model, without the abrupt layout changes that make adaptive interfaces hard to use [9], [10].

This paper makes three contributions:

- 1) A closed-loop architecture in which a BKT mastery estimate drives a PID governor whose output is a structural complexity budget, met by a deterministic layout planner and a generative model that writes the content, with schema validation and a deterministic fallback.
- 2) A computable structural complexity metric over a declarative component tree for linear-equation tasks, built from features linked in prior work to cognitive load and visual complexity.
- 3) An in-silico benchmark of 1,000 synthetic learners across four arms and ablations, including an analysis of BKT saturation after success runs and two remedies, with sensitivity to controller, metric and BKT parameters.

II. RELATED WORK

A. Learner modeling

Bayesian Knowledge Tracing (BKT) models each knowledge component as a two-state hidden Markov model with prior, learning, guess and slip parameters [1]. Extensions individualize parameters per learner [11], add forgetting [12], or contextualize guess and slip [13], and its fixed points and parameter constraints have been analyzed [14]. Logistic and deep alternatives exist [15]; BKT remains competitive when extended [12]. We use BKT because its state is a calibrated probability that can serve directly as a control variable.

B. Adaptive and generative interfaces

Model-based interface generation [16] adapts presentation to users, but adaptation has costs: unpredictable changes reduce usability [9], [10]. Recent GenUI systems let a large language

TABLE I
PRIOR APPROACHES AGAINST THE THREE PROPERTIES COMBINED HERE.
LEARNER MODEL MEANS A MODEL OF THE LEARNER'S KNOWLEDGE.

Approach	Learner model	Generated interface	Bounded by learner state
Tracing tutors [1], [2]	yes	no	no
Adaptive fading [8]	yes	no	fixed rule
Model-based UI [16]	no	yes	no
LLM-based GenUI [4], [5]	no	yes	no
This work	yes	yes	yes

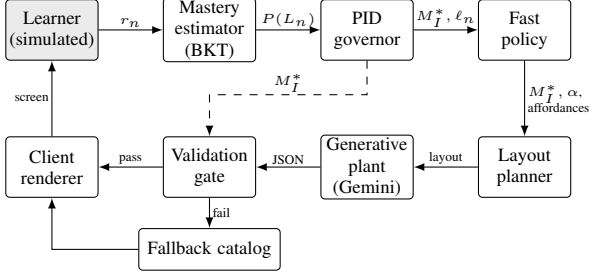


Fig. 1. The closed loop as deployed. The gate rechecks the budget (dashed); failing trees are replaced from the fallback catalog. The learner (shaded) is simulated here.

model produce the interface itself [4], [5]. To our knowledge, none of these systems bounds the generated interface by a learner model (Table I).

C. Cognitive load and scaffolding

Cognitive load theory [17] attributes intrinsic load to element interactivity [18]. Worked examples reduce load for algebra novices [19] and should fade into problem solving [7], [8]; concrete representations should fade into symbolic ones [20]. For linear equations, the balance model supports early understanding but breaks down with negative quantities [21], and learners hold persistent misconceptions about equality and operating on the unknown [22], [23].

D. Control and constrained generation

PID control with anti-windup and derivative filtering is standard practice [24], [25]. Constrained decoding guarantees well-formed model output [26], though schema-constrained output is not perfectly reliable in practice [27].

III. ARCHITECTURE

The loop has five stages (Fig. 1): a psychometric estimator, a complexity governor, a fast policy, a layout planner, and a generative plant whose output passes a validation gate before a deterministic client renders it.

A. Mastery estimation

After response $r_n \in \{0, 1\}$ the mastery estimate is updated by Bayes' rule and the learning transition [1]. Writing $p =$

$P(L_n)$, $g = P(G)$, $s = P(S)$ and $\hat{p} = P(L_n | r_n)$,

$$P(L_n | r_n=1) = \frac{p(1-s)}{p(1-s) + (1-p)g}, \quad (1)$$

$$P(L_n | r_n=0) = \frac{ps}{ps + (1-p)(1-g)}, \quad (2)$$

$$P(L_{n+1}) = \hat{p}(1 - P(F)) + (1 - \hat{p})P(T), \quad (3)$$

where $P(F) = 0$ gives standard BKT and $P(F) > 0$ adds forgetting [12]. Default parameters are $P(L_0) = 0.15$, $P(T) = 0.12$, $P(G) = 0.20$, $P(S) = 0.08$. They are design values, not fitted (sensitivity in Section V-C); they satisfy the bounds $P(G) < 0.3$, $P(S) < 0.1$ [1] and the non-degeneracy conditions $P(G) + P(S) < 1$, $P(T) < 1 - P(S)/(1 - P(G))$ [14].

B. Complexity governor

The setpoint is the conventional mastery threshold $P^* = 0.95$ [1], [28]. The tracking error is normalized on each side of the setpoint so that deficit and surplus have equal authority:

$$e_n = \begin{cases} (P^* - P(L_n))/P^* & P(L_n) \leq P^* \\ (P^* - P(L_n))/(1 - P^*) & P(L_n) > P^* \end{cases} \quad (4)$$

so that $e_n \in [-1, 1]$. The control effort is

$$u_n = K_p e_n + K_i S_n + K_d D_n, \quad (5)$$

with the clamped integral $S_n = \text{clamp}(S_{n-1} + e_n, -S_{\max}, S_{\max})$ for anti-windup [25] and the first-order filtered derivative $D_n = \gamma(e_n - e_{n-1}) + (1 - \gamma)D_{n-1}$ [24]. An inverted logistic maps effort to the complexity budget,

$$M_I^* = (1 + e^{ku_n})^{-1}, \quad (6)$$

which is quantized into four levels $\ell_n = \min(4, 1 + \lfloor 4M_I^* \rfloor)$. A hysteresis deadband Δ_h commits a new budget only when it differs from the last committed budget by at least Δ_h .

Gain selection: Classical tuning rules [29], [30] assume an identified plant response. In our benchmark the interface does not alter the simulated learner's responses (Section IV), so there is no plant response to identify. Instead, K_p , K_i , K_d , γ , $S_{\max}$ and Δ_h were selected by minimizing an ITAE-type tracking cost [31] plus a jitter penalty on a separate tuning cohort; all reported results use a disjoint test cohort. A random search of 200 samples over $K_p \in [0.25, 2.5]$, $K_i \in [0, 0.4]$, $K_d \in [0, 1]$, $\gamma \in [0.05, 1]$, $S_{\max} \in [1, 6]$ and $\Delta_h \in [0.02, 0.15]$ selected $K_p = 0.865$, $K_i = 0.377$, $K_d = 0.672$, $\gamma = 0.398$, $S_{\max} = 1.253$ and $\Delta_h = 0.141$. The cost surface is flat: the selected point scores 0.104 on the tuning cohort against 0.110 for hand-set initial values, which rank 22nd of 201. The logistic slope is $k = \ln 19/(K_p + K_i S_{\max}) = 2.202$, so that the effort range $\pm(K_p + K_i S_{\max})$ maps to $[0.05, 0.95]$.

C. Fast policy

A fast policy maps $(P(L_n), M_I^*, \ell_n, \text{recent errors})$ to discrete affordances: scaffolding mode (worked example, guided scaffold, open symbolic), whether to show the balance-scale manipulative, whether hints are enabled, and an instructional

density limit. The policy is deterministic and rule-based in every reported result. An open-weight decision model [32], evaluated zero-shot, took a median of 1.32 s per decision on a CPU and chose a worked example in all 200 test states, so it is not part of the evaluated system.

D. Layout planner, generative plant and validation gate

The planner turns the budget into a layout. The policy fixes the abstraction level α ; the planner then enumerates every admissible layout with 1–8 interactive elements, 0–3 hint tiers and nesting depth 1–4 (at most 168 candidates) and picks the one whose M_I is closest to M_I^* . It keeps a requested worked example when that stays within ε ; ties go to fewer elements, then shallower nesting. The plant, Gemini 3.7 Flash served through Vertex AI, receives this layout with the policy decision and the equation and writes the content as a JSON component tree. In the original design (*free* mode) the plant also chose the layout; we measure both (Table III). Nodes are drawn from a closed set of six components: CONCRETEBALANCESCALE, WORKED-SOLUTIONSTEP, SCAFFOLDEDOPERATIONPAD, SYMBOLICEQUATIONWORKSPACE, STEPPEDHINTACCORDION and CARTESIANVERIFICATIONPLANE. The client renders these components directly and never executes generated code. Each emission is parsed against the schema; if parsing fails, or if $|M_I - M_I^*| > \varepsilon$ with $\varepsilon = 0.05$, a precompiled fallback tree nearest to the budget is rendered instead. A web client renders the trees for inspection (Fig. 2); the benchmark does not use it.

E. Structural complexity metric

For a tree with ρ interactive elements, abstraction level α and nesting depth δ ,

$$M_I = w_1 \frac{\rho - 1}{7} + w_2 \frac{\alpha - 1}{3} + w_3 \frac{\delta - 1}{3}, \quad (7)$$

with $\rho \in [1, 8]$ and $\alpha, \delta \in [1, 4]$. Each feature has a basis in prior work: element interactivity raises load [18] and display density is a classic complexity measure [33]; the abstraction ladder from balance scale to operation pad to symbolic workspace follows concreteness fading [20]; and deeper structures cost users more [34]. Placing the Cartesian plane at the top of the ladder is our design choice. With no data to fit weights, we use equal weights $w_i = 1/3$, which are robust in that setting [35], and test sensitivity to the weights (Section V). M_I is a structural proxy; it has not been validated against measured cognitive load.

IV. METHOD

A. Synthetic learners

Simulated learners are an established way to test instructional designs before human studies [36], [37], and BKT-generated data have been used to evaluate mastery assessment [38]. Each learner is a two-state HMM whose parameters are drawn uniformly from archetype ranges (Table II). The ranges for the novice and guesser archetypes exceed the tighter

TABLE II
LEARNER ARCHETYPES (UNIFORM PARAMETER RANGES).

Archetype	$P(L_0)$	$P(T)$	$P(G)$	$P(S)$
Struggling novice	.02–.08	.03–.07	.10–.20	.20–.30
Inconsistent guesser	.05–.20	.10–.20	.15–.40	.08–.30
Fast master	.15–.30	.30–.40	.15–.25	.01–.03

TABLE III
GEMINI 3.7 FLASH, 300 CALLS PER MODE.

	Free	Guided
Transport failures (HTTP error, timeout)	6	4
Content-malformed, of completed calls	3.4%	0%
Off budget, of valid trees	2.8%	0%
Within budget, of all calls	92.0%	98.7%
Median $ M_I - M_I^* $ of valid trees	0.012	0.007
Median thinking tokens	696	0
Latency of completed calls, median / p95	7.7 / 18.6 s	2.7 / 5.1 s

bounds of [1] by design to represent noisy learners, while respecting the non-degeneracy conditions of [14]. The cohort has 1,000 learners (300/400/300) with 40 items each. Responses are sampled once per learner and replayed unchanged through every arm (paired design). The interface therefore does not influence the simulated responses: the benchmark measures controller behavior, not learning.

B. Arms

(1) *Static*: a fixed level-2 interface. (2) *Unconstrained GenUI*: the plant chooses complexity from a noisy reading of mastery with no budget check. (3) *Rule-based*: step up after three consecutive correct responses and down after two consecutive errors, an up-down rule in the spirit of [39]. (4) *Closed loop*: the full architecture. Ablations remove the derivative filter, remove anti-windup, and add a one-level slew limit.

C. Plant model in the benchmark

The benchmark replaces the foundation model with a surrogate that compiles the planner’s layout and injects faults: with probability 0.15 it drifts off budget, and with probability 0.03 it emits malformed output (truncated JSON, unknown component, out-of-range field, depth overflow, injected field). These rates were set before the real model was measured. The unconstrained arm uses the same surrogate without the budget check.

We measured the real model in free mode, which tests the model alone, and in guided mode, the deployed configuration with the planner. Each mode received 300 requests (JSON mode without a response schema, thinking level low, temperature 0.4, 30 s timeout) with budgets uniform on $[0.02, 0.98]$, scored by the same gate (Table III).

In free mode, ten of 294 completed calls lacked an interactive element and eight of 284 valid trees missed the budget, most often at level 1; in guided mode all failures were in transport. The surrogate’s 3% malformed rate thus matches

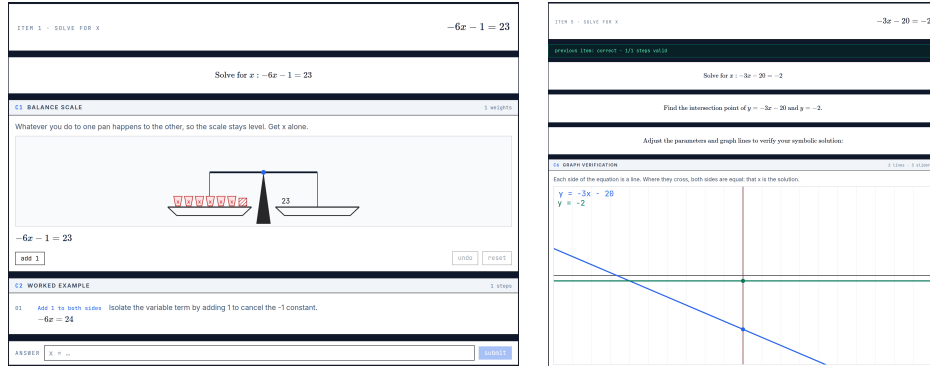


Fig. 2. Screens written by Gemini 3.7 Flash in guided mode; both passed the gate. Left: level 1, balance scale with a worked step ($M_I^* = 0.091$, $M_I = 0.111$). Right: level 4, graph verification ($M_I^* = 0.903$, $M_I = 0.905$).

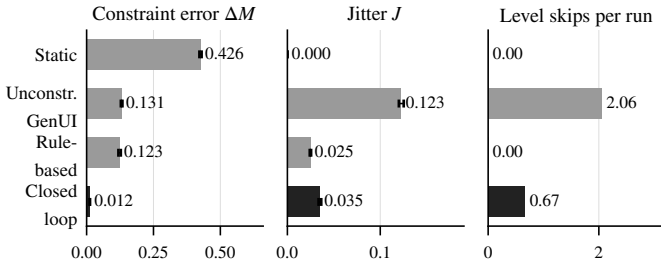


Fig. 3. Main comparison on the test cohort (means, 95% CIs); closed loop dark.

free mode, and its 15% drift rate is a stress setting. Free mode’s latency comes largely from hidden reasoning (our reading).

D. Metrics

Constraint error $\Delta M = |M_I - M_{\text{ref}}^*|$ is measured against the budget the closed-loop governor commits for the same responses, so that all arms share one reference; “within ε ” is the share of steps with $\Delta M \leq 0.05$. Interface jitter is $J = \frac{1}{N-1} \sum_n |M_I(n) - M_I(n-1)|$, and a skip is a change of two or more levels between consecutive items. Jitter operationalizes the stability concern of [9]; the formula is ours. Recovery time is the number of items after a streak of five or more errors until the level returns to its pre-streak value. Fallback rate is the share of steps rendered from the fallback catalog. Because the simulator knows the learner’s latent state, we also report *overload* (level 3 or 4 shown while the skill is not known), *underload* (level 1 or 2 shown while it is known) and an ITAE-weighted mismatch $\sum_n (n+1)m_n / \sum_n (n+1)$ [31], where m_n marks either case.

V. RESULTS

All numbers come from one benchmark run on the test cohort (seed 20261003; tuning seed 20261004). Intervals are normal-approximation 95% confidence intervals over the 1,000 learners.

A. Main comparison

The closed loop keeps the rendered interface within 0.012 of its budget on average (Table IV, Fig. 3), against 0.131

and 0.123 for unconstrained GenUI and the rule-based arm. Every rendered tree is within ε , partly by construction: the generator’s own output met the budget on 84.6% of steps, and the gate replaced the other 15.4% from the fallback catalog. The compliance is thus that of governor, planner and gate together, not of the governor alone. The closed loop’s jitter (0.035) is 3.5 times lower than that of unconstrained GenUI (0.123), which also skips levels 2.06 times per run against 0.67. The rule-based arm is smoother still (0.025, no skips), because it moves one level at a time. Against the latent learner state, the picture is mixed. Unconstrained GenUI has the lowest ITAE mismatch (0.043 against 0.073) and the least underload (5.4% of steps against 12.1%), but shows an abstract interface to a learner who does not know the skill twelve times as often (2.4% against 0.2%). The rule-based arm underloads on 24.5% of steps and the static arm on 80%. After a streak of five or more errors that lowered the level, the closed loop returned to its pre-streak level within the horizon in 34 of 51 cases, after a median of 5 items; the rule-based arm did so in 9 of 19 cases, after a median of 12.

B. Ablations

Removing the derivative filter raises jitter from 0.035 to 0.044 and nearly doubles level skips (1.19 per run). Removing anti-windup lets the integral grow without bound: underload rises from 12.1% to 21.7%, only 84% of learners reach level 4 against 95%, and median recovery after an error streak grows from 5 to 14 items (only 4 recoveries within the horizon). In a scripted probe of prolonged struggle, the unclamped integral reached 22.9 (clamp 1.25) and the interface never reached level 4. A one-level slew limit removes nearly all skips (0.02 per run) and leaves the ground-truth measures unchanged.

C. Sensitivity

Changing each controller parameter by $\pm 50\%$ raised the test cost (ITAE + J) from 0.108 to at most 0.118, except $1.5\Delta_h$, which lowered it to 0.104 but more than doubled level skips. For the metric weights, the closed loop’s ΔM stays between 0.007 and 0.017 under equal weights, the initial 0.40/0.35/0.25 split and three single-feature-heavy splits

TABLE IV

MAIN RESULTS AND ABLATIONS ON THE TEST COHORT (1,000 LEARNERS, 40 ITEMS EACH). ΔM AND “WITHIN ϵ ” ARE MEASURED AGAINST THE CLOSED-LOOP REFERENCE BUDGET; EVERY CLOSED-LOOP VARIANT MEETS ITS OWN BUDGET AT EVERY STEP. OVERLOAD, UNDERLOAD AND ITAE USE THE LATENT LEARNER STATE AND EXIST ONLY IN SIMULATION. FALLBACK RATES FOLLOW FROM THE SURROGATE’S FAULT RATES; ITS DRIFT RATE IS ABOUT FIVE TIMES THAT MEASURED ON THE REAL MODEL.

Arm	ΔM (95% CI)	Within ϵ	Jitter J	Skips/run	Overload	Underload	ITAE	Fallback
Static	0.426 (0.422–0.430)	0.004	0.000	0.00	0.000	0.800	0.892	–
Unconstrained GenUI	0.131 (0.128–0.134)	0.321	0.123	2.06	0.024	0.054	0.043	0.030
Rule-based	0.123 (0.119–0.128)	0.596	0.025	0.00	0.000	0.245	0.180	–
Closed loop	0.0122 (0.0119–0.0124)	1.000	0.035	0.67	0.002	0.121	0.073	0.154
no derivative filter	0.031 (0.030–0.033)	0.811	0.044	1.19	0.002	0.124	0.076	0.156
no anti-windup	0.107 (0.102–0.112)	0.549	0.024	0.42	0.001	0.217	0.174	0.159
one-level slew limit	0.025 (0.023–0.026)	0.875	0.034	0.02	0.002	0.121	0.073	0.153

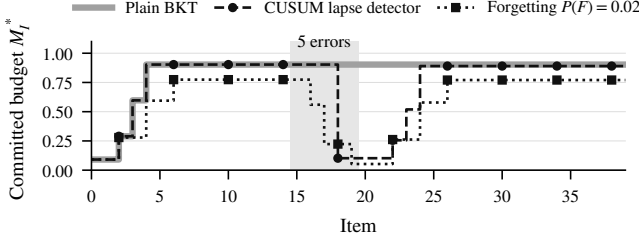


Fig. 4. Scripted probe: 15 correct responses, five errors, then correct again. With plain BKT the estimate is saturated and the budget never moves.

(0.6/0.2/0.2). Dropping abstraction or depth roughly doubles it (0.023 and 0.022) while keeping more than 99% of steps within ϵ . Dropping element density breaks the loop: fallbacks rise to 84% and only 18% of screens are within ϵ . Changing each BKT parameter by $\pm 50\%$ kept every step within ϵ , ΔM between 0.010 and 0.017, jitter between 0.034 and 0.041 and fallbacks at 15–16%. The parameters mainly shift when the interface escalates: $P(T)$ and $P(G)$ move ITAE between 0.048 and 0.115, and skips rise to at most 1.20 per run.

D. BKT saturation and lapse handling

Under correct responses the BKT estimate converges to a stable fixed point at one [14]. With the default parameters the estimate exceeds $1 - 10^{-6}$ after ten correct responses and five subsequent errors lower it only to about 0.98; after 24 correct responses it equals 1.0 in double precision and no further evidence can lower it. The interface therefore stays at its highest level through a streak of errors. We compare two remedies: forgetting, $P(F) > 0$ [12], and a CUSUM detector [40] on the log-likelihood ratio between “not known” and the tracker’s prediction, which resets the estimate to 0.5 when the sum exceeds $h = 6.5$. Because it accumulates only when a confident estimate is contradicted, isolated slips are absorbed. The tracker with forgetting uses $P(F) = 0.02$.

In a scripted probe with fifteen correct responses followed by five errors (Fig. 4), the plain closed loop never lowers the interface; the CUSUM variant drops a level 3 items after the streak begins and forgetting after 1. To test this at scale, a second cohort of 1,000 learners can forget the skill with probability 0.02–0.06 per item. There the plain closed loop shows

an abstract interface to a learner who does not know the skill on 7.6% of steps. The CUSUM detector lowers this to 5.6% (–25%) at the cost of higher jitter (0.052 against 0.042) and an almost unchanged ITAE mismatch (0.189 against 0.184). Forgetting lowers overload further, to 3.0%, but underloads on 25.3% of steps against 12.8%, raises ITAE to 0.273 and lets only 80% of learners reach level 4. Neither remedy dominates. Weighting recent responses more heavily [41] is a third option we leave for future work.

VI. DISCUSSION AND LIMITATIONS

The benchmark tests whether the controller does what it is designed to do; it does not test whether learners benefit. Because simulated responses do not depend on the interface, the loop is closed only through the mastery estimate, not through the learner. M_I is a structural proxy that has not been validated against measured load, for which instruments exist [42]. BKT parameters are design values rather than fits to tutor data; $\pm 50\%$ changes leave tracking intact but shift when the interface escalates. Long error streaks may reflect misconceptions rather than slips [23]; the controller does not distinguish these. The benchmark plant is a surrogate. Its malformed rate matches and its drift rate exceeds those measured on the real model, but the real-model measurement covers single calls, not full learner trajectories. Its latency (median 7.7 s free, 2.7 s guided) calls for prefetching or caching. A learned fast policy would need real interaction data; simulator data would only imitate the rule policy.

VII. CONCLUSION

We coupled a learner model to a generative interface through a PID governor, a fast policy, a layout planner and a validation gate, with the generative model writing content into a fixed layout, and tested the loop in simulation. The governor’s budget is met at every step, with the gate replacing the 15.4% of generated trees that miss it, and the interface changes 3.5 times less than an unconstrained generator while rarely showing abstract screens to simulated learners who have not mastered the skill. Anti-windup and the derivative filter each matter, and a slew limit removes level skips. Standard BKT stops reacting after long success runs; a CUSUM lapse detector restores the reaction at a modest cost in jitter, while

forgetting overcorrects. In 300 single real calls, 98.7% of screens met the budget with the planner and 92% without it. The next steps are running full trajectories with the real model, a learned fast policy trained on real interaction data, and a study with learners.

CODE AND DATA AVAILABILITY

The engine, simulation harness, benchmark results and the sources of this manuscript are openly available at <https://github.com/omanandswami2005/closed-loop-genui-research-implementation>.

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